# Phase 1 Report: DOA-Sector Inference with LIF-Inspired Edge Processing

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**Abstract**

This report documents Phase 1 of an embedded-AI project focused on lightweight spike-based inference for edge sensing. A complete Python reference prototype was developed for direction-of-arrival (DOA) sector classification using a physics-based uniform linear array (ULA) signal model. The processing chain includes covariance estimation, beamspace feature extraction, feature normalization, nearest-centroid baseline classification, rate-based spike encoding, and leaky-integrate-and-fire (LIF)-inspired temporal inference. The nearest-centroid baseline achieved an overall test accuracy of 0.869048. A fixed LIF configuration achieved 0.825000 accuracy with a mean stable-decision latency of 6.423810 time steps. A later LIF parameter sweep improved the best LIF accuracy to 0.841667 at $T = 40$ and leak = 0.99. These results demonstrate a clear and controllable accuracy–latency trade-off, which is central to the embedded AI motivation of the project.

## Contents

**1 Purpose and Scope** **2**

**2 Signal Model and Dataset** **2**

**3 Beamspace Features** **3**

**4 Nearest-Centroid Baseline** **3**

**5 Spike Encoding and LIF-Inspired Inference** **4**

**6 Accuracy and Latency Analysis** **4**

**7 Run 005 Parameter Sweep** **6**

**8 Engineering Interpretation** **7**

**9 Conclusion** **7**

**10 References** **8**

# 1 Purpose and Scope

The purpose of Phase 1 is to establish a complete and reproducible Python reference model for DOA-sector inference using a physics-based sensing model and a hardware-aware spike-based inference path. This phase is not intended to introduce a new high-resolution DOA algorithm. Instead, it provides an engineering baseline that can later be refactored toward embedded execution, PYNQ deployment, or Vitis HLS acceleration.

The completed prototype connects three technical layers:

1. array signal processing through a ULA signal model and covariance-derived beamspace features,
2. machine-learning baseline evaluation through nearest-centroid classification,
3. sparse temporal inference through rate-based spike encoding and LIF-inspired evidence accumulation.

# 2 Signal Model and Dataset

The Phase 1 dataset is generated using a narrowband single-source ULA model. For each source angle $\theta$ and SNR value, the received array snapshot matrix is modeled as

$$X = a(\theta)s + N, \tag{1}$$

where $a(\theta)$ is the ULA steering vector, $s$ is a complex narrowband source signal, and $N$ is circular complex additive white Gaussian noise. The sample covariance matrix is computed as

$$R_{xx} = \frac{1}{M} X X^H, \tag{2}$$

where $M = 128$ snapshots.

A compact beamspace feature vector is extracted from the covariance matrix by evaluating spatial energy over a fixed angular grid. The features are normalized using training-set statistics before classification or spike encoding.

Table 1: Phase 1 Dataset Configuration

| Parameter | Value |
|---|---|
| Array type | Uniform Linear Array |
| Number of array elements | 8 |
| Snapshots per sample | 128 |
| Element spacing | $\lambda/2$ |
| Source count | 1 |
| DOA range | $-60^\circ$ to $+60^\circ$ |
| Number of DOA sectors | 4 |
| SNR values | $-10, -5, 0, 5, 10, 15, 20$ dB |
| Samples per sector per SNR | 100 |
| Total samples | 2800 |
| Train/test split | 70/30 stratified |
| Test samples | 840 |

## 3 Beamspace Features

Beamspace features provide a compact representation of angular information. The objective is not to estimate the exact continuous DOA angle, but to classify the received signal into one of four angular sectors. A beamspace representation is appropriate for this task because it reduces the complex covariance structure into a small real-valued feature vector that captures spatial energy distribution.

Figure 1 shows an example beamspace response generated during the reference run.

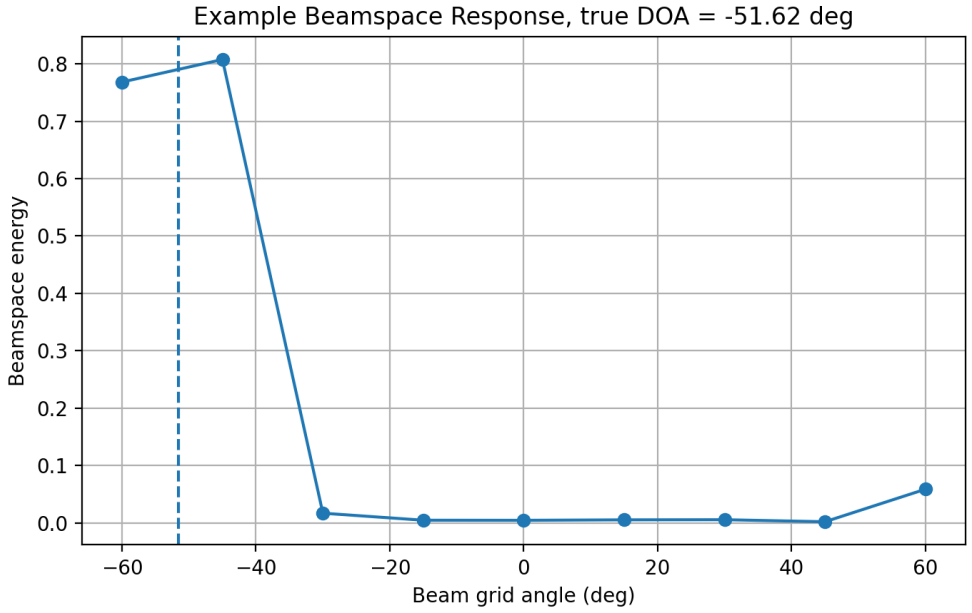



Figure 1: Example beamspace response from the Phase 1 ULA simulation.

## 4 Nearest-Centroid Baseline

The non-spiking reference classifier is a nearest-centroid classifier trained on the normalized beamspace features. This classifier is deterministic, interpretable, and lightweight. Its purpose is to establish the amount of DOA-sector information available in the compact feature vector before converting the features into spike trains.

The nearest-centroid baseline achieved

$$\text{Accuracy}_{\text{baseline}} = 0.869048. \tag{3}$$

Table 2: Nearest-Centroid Accuracy Versus SNR

| SNR (dB) | Accuracy |
|---|---|
| -10 | 0.758333 |
| -5 | 0.908333 |
| 0 | 0.891667 |
| 5 | 0.858333 |
| 10 | 0.866667 |
| 15 | 0.925000 |
| 20 | 0.875000 |

## 5 Spike Encoding and LIF-Inspired Inference

Normalized beamspace features are converted into binary spike trains using rate-based encoding. A larger normalized feature value corresponds to a higher probability of spike emission across the time window. This representation is sparse and temporal, making it a useful abstraction for later hardware-oriented experiments.

The LIF-inspired classifier is implemented as a class-level leaky membrane accumulator. It is not a biologically detailed SNN. Instead, it is a hardware-facing approximation in which each class accumulates spike-driven evidence over time. The membrane state can be interpreted as

$$v_c[t] = \alpha v_c[t-1] + i_c[t], \tag{4}$$

where $v_c[t]$ is the membrane value for class $c$, $\alpha$ is the leak factor, and $i_c[t]$ is the spike-driven input current at time step $t$.

The fixed Run 003 configuration used

$$T = 20, \qquad \alpha = 0.97. \tag{5}$$

It achieved

$$\text{Accuracy}_{\text{LIF, fixed}} = 0.825000, \tag{6}$$

with mean stable-decision latency

$$\bar{L} = 6.423810 \text{ time steps}. \tag{7}$$

## 6 Accuracy and Latency Analysis

Run 004 generated report-ready figures comparing the nearest-centroid baseline and the LIF-inspired classifier across the same SNR grid. Figure 2 compares accuracy versus SNR. Figure 3 shows mean stable-decision latency versus SNR.

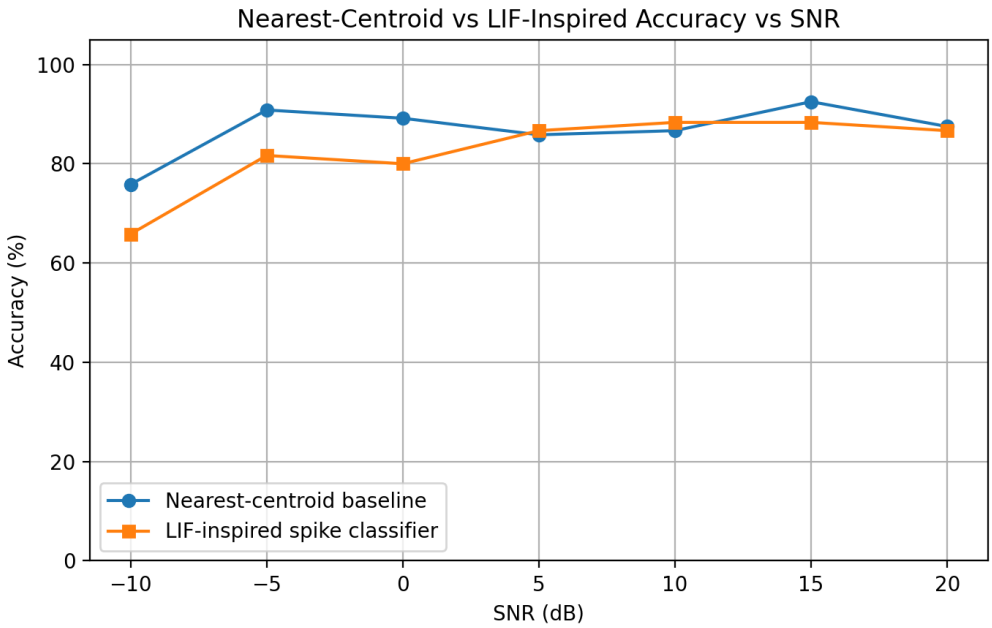



Figure 2: Accuracy comparison between the nearest-centroid baseline and the LIF-inspired classifier.

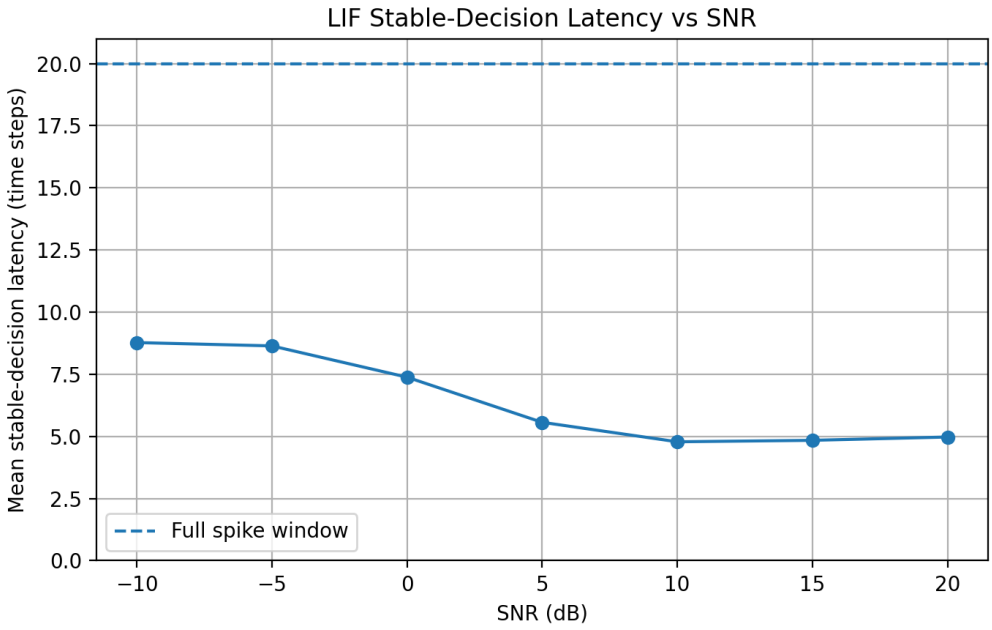



Figure 3: Mean stable-decision latency of the LIF-inspired classifier versus SNR.

The latency trend supports the edge-inference interpretation. When the signal quality is higher, the LIF decision generally stabilizes earlier. This suggests that temporal inference can support early-exit or adaptive-latency decision logic in later embedded implementations.

# 7 Run 005 Parameter Sweep

Run 005 evaluated the sensitivity of the LIF-inspired classifier to spike-window length and membrane leak factor. The sweep grid was

$$T \in \{10, 20, 30, 40\}, \qquad \alpha \in \{0.90, 0.95, 0.97, 0.99\}. \tag{8}$$

This created 16 configurations.

Table 3: Run 005 Key Results

| Case | $T$ | Leak | Accuracy | Mean latency |
|---|---|---|---|---|
| Best LIF accuracy | 40 | 0.99 | 0.841667 | 8.396429 |
| Lowest latency | 10 | 0.99 | 0.776190 | 3.782143 |
| Fixed Run 003 | 20 | 0.97 | 0.825000 | 6.423810 |
| Non-spiking baseline | – | – | 0.869048 | – |

Table 4: Full Run 005 LIF Parameter Sweep

| $T$ | Leak | Accuracy | Mean latency |
|---|---|---|---|
| 10 | 0.90 | 0.773810 | 4.280952 |
| 10 | 0.95 | 0.778571 | 4.011905 |
| 10 | 0.97 | 0.779762 | 3.914286 |
| 10 | 0.99 | 0.776190 | 3.782143 |
| 20 | 0.90 | 0.809524 | 7.732143 |
| 20 | 0.95 | 0.816667 | 6.579762 |
| 20 | 0.97 | 0.819048 | 6.340476 |
| 20 | 0.99 | 0.816667 | 5.753571 |
| 30 | 0.90 | 0.802381 | 10.946429 |
| 30 | 0.95 | 0.829762 | 8.870238 |
| 30 | 0.97 | 0.838095 | 8.126190 |
| 30 | 0.99 | 0.832143 | 7.352381 |
| 40 | 0.90 | 0.801190 | 14.945238 |
| 40 | 0.95 | 0.822619 | 10.778571 |
| 40 | 0.97 | 0.830952 | 9.697619 |
| 40 | 0.99 | 0.841667 | 8.396429 |

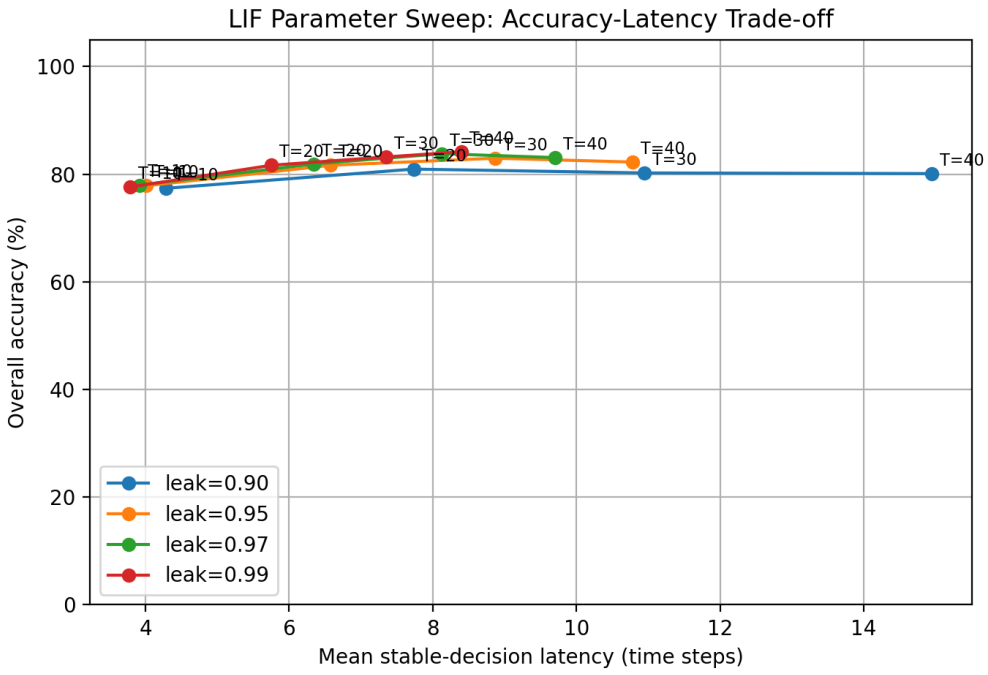



Figure 4: Run 005 LIF accuracy–latency trade-off across time-window and leak configurations.

## 8 Engineering Interpretation

The parameter sweep confirms a clear accuracy–latency design trade-off. Increasing the spike time window generally improves classification accuracy because the rate-coded representation has more time steps to approximate the original continuous feature vector. However, a longer time window can increase temporal budget and stable-decision latency.

The leak factor controls evidence retention. Higher leak values preserve accumulated information better. In this prototype, the best LIF accuracy occurs at $T = 40$ and leak $= 0.99$. The best LIF result improves over the fixed Run 003 configuration but remains below the continuous nearest-centroid baseline. This result is expected and useful: spike-based inference trades some continuous-valued accuracy for sparse temporal processing and explicit latency control.

## 9 Conclusion

The Phase 1 Python prototype successfully establishes an end-to-end DOA-sector inference pipeline based on a physics-derived ULA signal model. The nearest-centroid classifier demonstrates that compact beamspace features provide strong sector-level DOA information, achieving 86.9% test accuracy.

The LIF-inspired classifier demonstrates that the same normalized features can be converted into sparse temporal spike trains and classified using a hardware-facing leaky evidence accumulator. The fixed LIF configuration achieves 82.5% accuracy, while the Run 005 parameter sweep improves the best LIF result to 84.17%.

The Phase 1 result is therefore a valid embedded-AI reference baseline. The natural next step is to convert the Python reference model into a more formal course report and then extend the

project toward richer radar/DOA case studies or embedded implementation.

## 10 References

## References


[1] H. L. Van Trees, *Optimum Array Processing*. Wiley, 2002.

[2] S. Haykin, *Adaptive Filter Theory*. Pearson, 2014.

[3] K. Roy, A. Jaiswal, and P. Panda, "Towards spike-based machine intelligence with neuromorphic computing," *Nature*, vol. 575, pp. 607–617, 2019.

[4] W. Maass, "Networks of spiking neurons: The third generation of neural network models," *Neural Networks*, vol. 10, no. 9, pp. 1659–1671, 1997.